

Confidence Intervals Worksheet

Estimating Battery Performance & Reliability in Electric Vehicles

Part A – Conceptual Understanding

Q1. Define the following terms with your own words and an EV example:

a. Confidence Interval (CI)

A confidence interval is a range of values that likely contains the true average value of something we're studying.

EV Example: If we test the battery range of 25 electric vehicles and find a 95% CI of [280, 320] miles, it means we are fairly confident that the *average battery range of all EVs* falls within this range.

b. Confidence Level (CL)

The confidence level is how sure we are that the confidence interval actually includes the true average.

EV Example: A 95% confidence level means that if we repeated our EV battery test many times, 95% of the calculated intervals would include the true average battery range.

c. Margin of Error (ME)

The margin of error shows how much we expect our sample mean to differ from the true mean.

EV Example: If the average battery range from our sample is 300 miles with a margin of error of ± 20 miles, then we estimate the true mean is between 280 and 320 miles.

Q2. Choose the correct answer ():

A 95% CI means:

The sample mean is always within this range

The true mean has a 95% probability of falling within this range

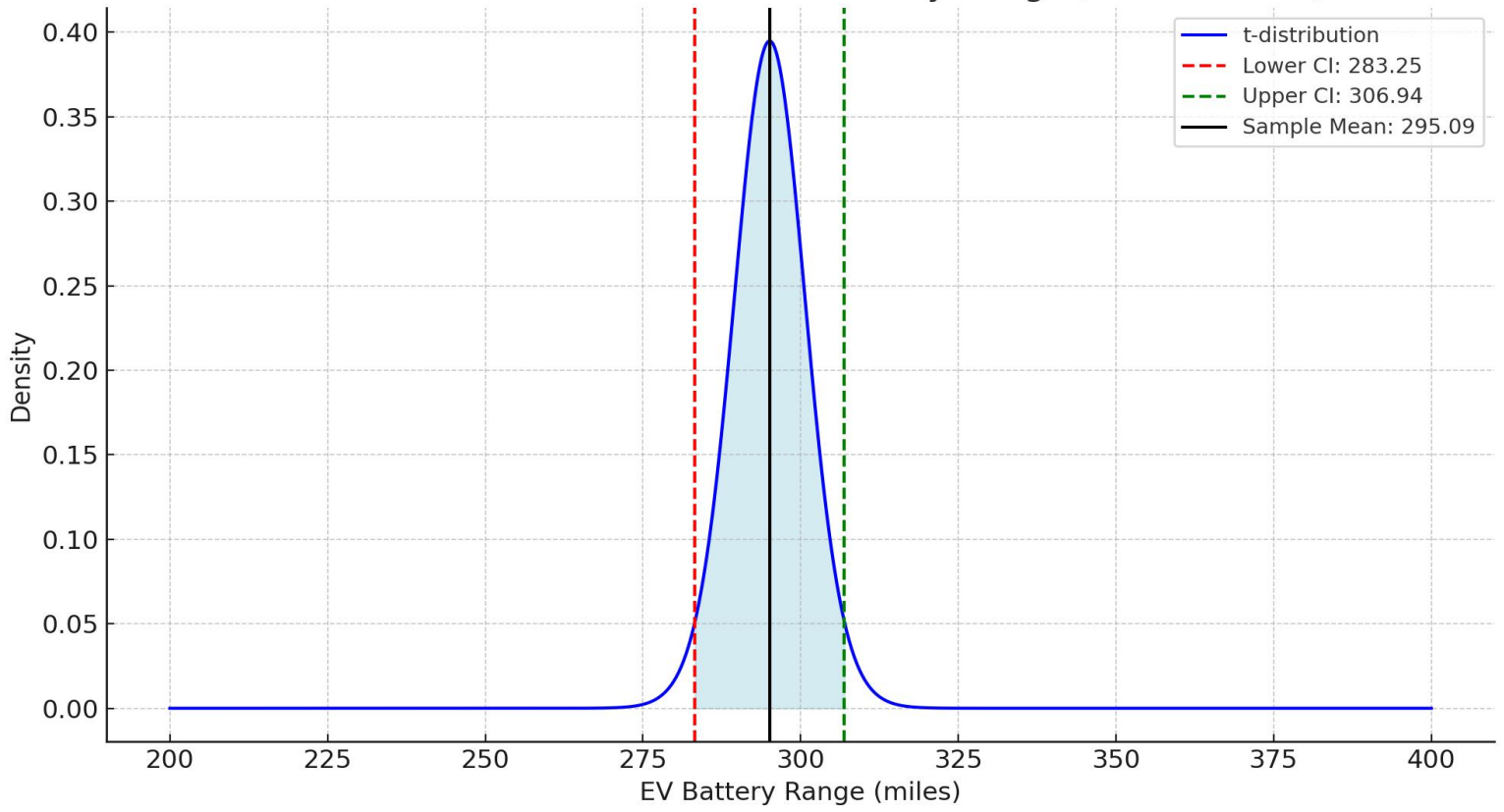
It's about long-run coverage over repeated samples

The standard deviation is 95%

Explanation:

A 95% confidence interval means that if we took many samples and built a CI from each one, 95% of those intervals would contain the true population mean. It does **not** mean there's a 95% chance the true mean is in one specific interval.

95% Confidence Interval for EV Battery Range (t-distribution)



Q3. When should you use a t-distribution over a Z-distribution? Explain with an EV battery example.

When to Use:

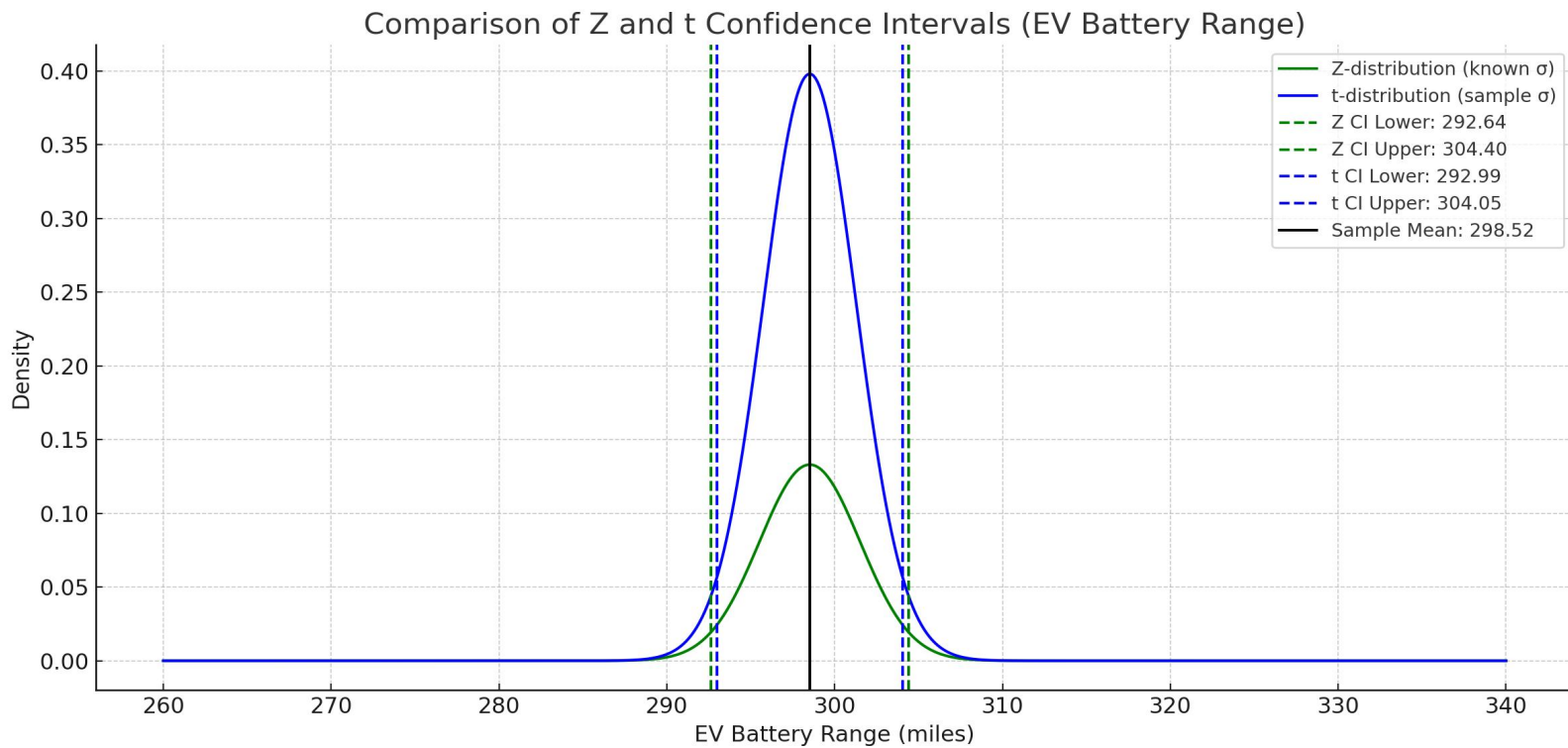
- Use **t-distribution** when:
 - The sample size is small (typically < 30)
 - The population standard deviation is unknown
- Use **Z-distribution** when:
 - The sample size is large
 - The population standard deviation is known

EV Example:

We are testing the battery range of a small sample (25 EVs). Since you're unlikely to know the **true population standard deviation**, we would use a **t-distribution** to calculate your confidence interval.

Visualization Using Python

Python Code for t vs. Z Confidence Interval Comparison:



Python Code

```
import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt

# Sample data: EV battery ranges (simulated)
np.random.seed(42)
sample_size = 25
sample = np.random.normal(loc=300, scale=15, size=sample_size)
sample_mean = np.mean(sample)
sample_std = np.std(sample, ddof=1)

# Confidence intervals
confidence = 0.95
alpha = 1 - confidence

# t-distribution CI
t_crit = stats.t.ppf(1 - alpha/2, df=sample_size-1)
margin_error_t = t_crit * (sample_std / np.sqrt(sample_size))
ci_t = (sample_mean - margin_error_t, sample_mean + margin_error_t)

# Z-distribution CI (assuming known population std = 15)
z_crit = stats.norm.ppf(1 - alpha/2)
margin_error_z = z_crit * (15 / np.sqrt(sample_size))
ci_z = (sample_mean - margin_error_z, sample_mean + margin_error_z)

# Plotting
x = np.linspace(sample_mean - 4*sample_std, sample_mean + 4*sample_std, 1000)
y_t = stats.t.pdf((x - sample_mean)/(sample_std / np.sqrt(sample_size)), df=sample_size-1)
y_z = stats.norm.pdf(x, loc=sample_mean, scale=15/np.sqrt(sample_size))

plt.figure(figsize=(10, 6))
plt.plot(x, y_t, label='t-distribution', color='blue')
plt.plot(x, y_z, label='z-distribution', color='green')
plt.axvline(ci_t[0], color='blue', linestyle='--', label='CI t-lower')
plt.axvline(ci_t[1], color='blue', linestyle='--', label='CI t-upper')
plt.axvline(ci_z[0], color='green', linestyle='--', label='CI z-lower')
plt.axvline(ci_z[1], color='green', linestyle='--', label='CI z-upper')
plt.axvline(sample_mean, color='black', linestyle='--', label='Sample Mean')

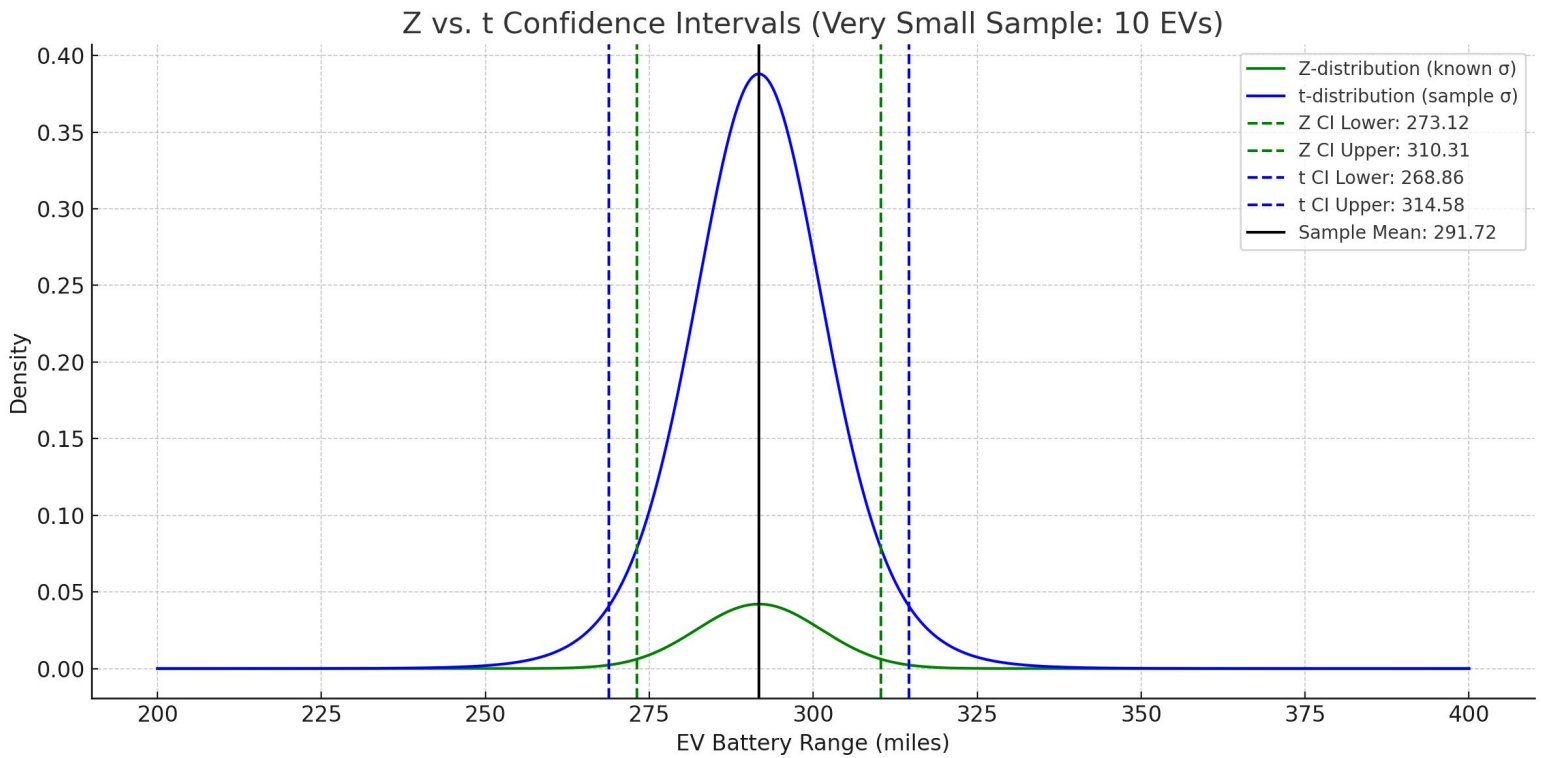
plt.title("EV Battery Range: t vs. Z Confidence Interval")
plt.xlabel("Battery Range (miles)")
plt.ylabel("Probability Density")
plt.legend()
plt.grid(True)
plt.show()
```

Interpretation:

Green Curve (Z-distribution): Used when the population standard deviation is known and the sample size is large. It's narrower due to less uncertainty.

Blue Curve (t-distribution): Used when the standard deviation is estimated from the sample. It's wider, especially for small samples.

Dashed Lines: Show the CI range. Blue (t) is wider than green (Z), reflecting higher uncertainty.



Confidence Intervals in Electric Vehicle Analysis

Question 4 – Confidence Interval for Average EV Range

Concept: Problem Setup

You tested 10 electric vehicles and recorded the following ranges (in km):

EV Ranges: [285, 290, 295, 270, 275, 300, 310, 280, 295, 288]

We assume the population standard deviation is unknown, so we will use the t -distribution to calculate the 95% confidence interval.

Step 1: Compute Sample Statistics

$$\bar{x} = \frac{1}{10} \sum x_i = 288.8 \text{ km}$$
$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = 12.74 \text{ km}$$

Concept: Degrees of Freedom

Since $n = 10$, the degrees of freedom are $df = n - 1 = 9$. We find the critical t -value for a 95% confidence level and 9 degrees of freedom:

$$t^* = 2.262$$

Step 2: Calculate Margin of Error and CI

$$\text{Margin of Error} = t^* \cdot \frac{s}{\sqrt{n}} = 2.262 \cdot \frac{12.74}{\sqrt{10}} \approx 8.56 \text{ km}$$

$$\text{Confidence Interval} = \bar{x} \pm \text{ME} = 288.8 \pm 8.56$$

$$\Rightarrow [280.24 \text{ km}, 297.36 \text{ km}]$$

Visual Interpretation

Observation

A plot shows:

- A black line at the sample mean (288.8 km)
- A shaded blue region from 280.24 to 297.36 km (the confidence interval)
- Red and green dashed lines at the CI boundaries
- Blue dots representing individual EV range data

Insight: The 95% CI gives us a range where we expect the true average EV range to fall with 95% confidence.

Extended Analysis – 20 Vehicles

Updated Sample

We test 10 additional EVs:

[292, 298, 305, 287, 290, 299, 284, 293, 296, 291]

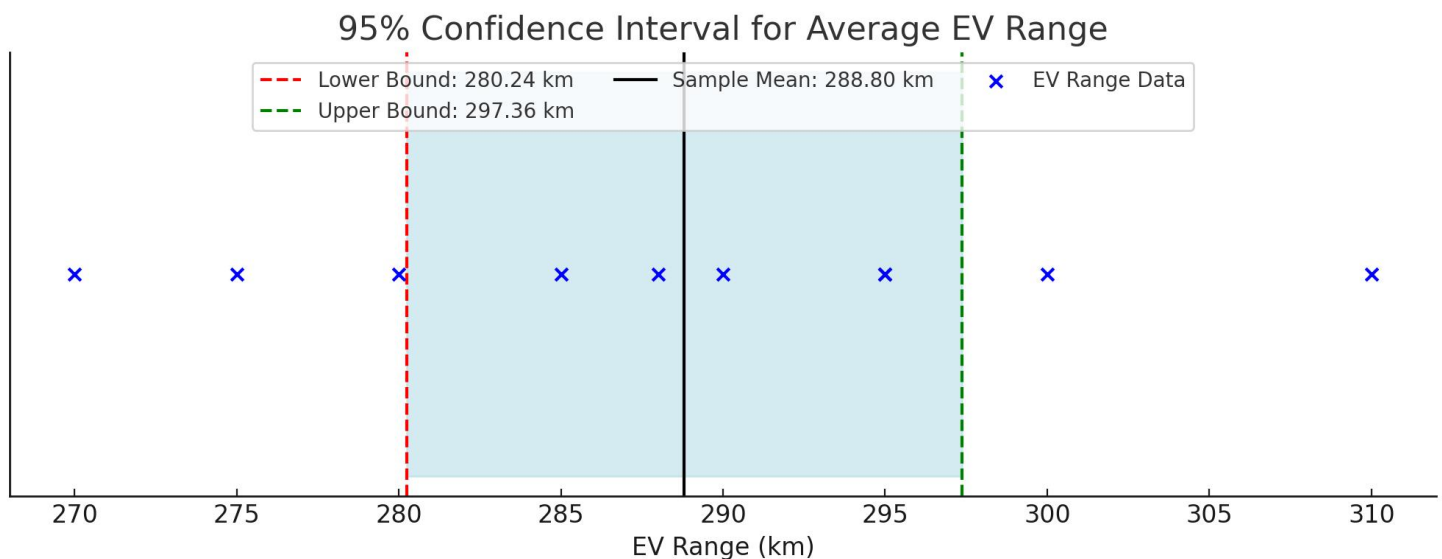
$$\bar{x}_{20} = 291.15 \text{ km}, \quad s_{20} = 11.28 \text{ km}, \quad t_{df=19}^* = 2.093$$

$$ME_{20} = 2.093 \cdot \frac{11.28}{\sqrt{20}} \approx 4.48 \text{ km}$$

$$CI_{20} = [286.67 \text{ km}, 295.63 \text{ km}]$$

Observation

With more vehicles tested ($n = 20$), the margin of error shrinks and the CI narrows. This makes the estimate of average EV range more precise.



Here's the visualization of the 95% confidence interval for the average EV range:

Red dashed line: Lower bound (≈ 280.24 km)

Green dashed line: Upper bound (≈ 297.36 km)

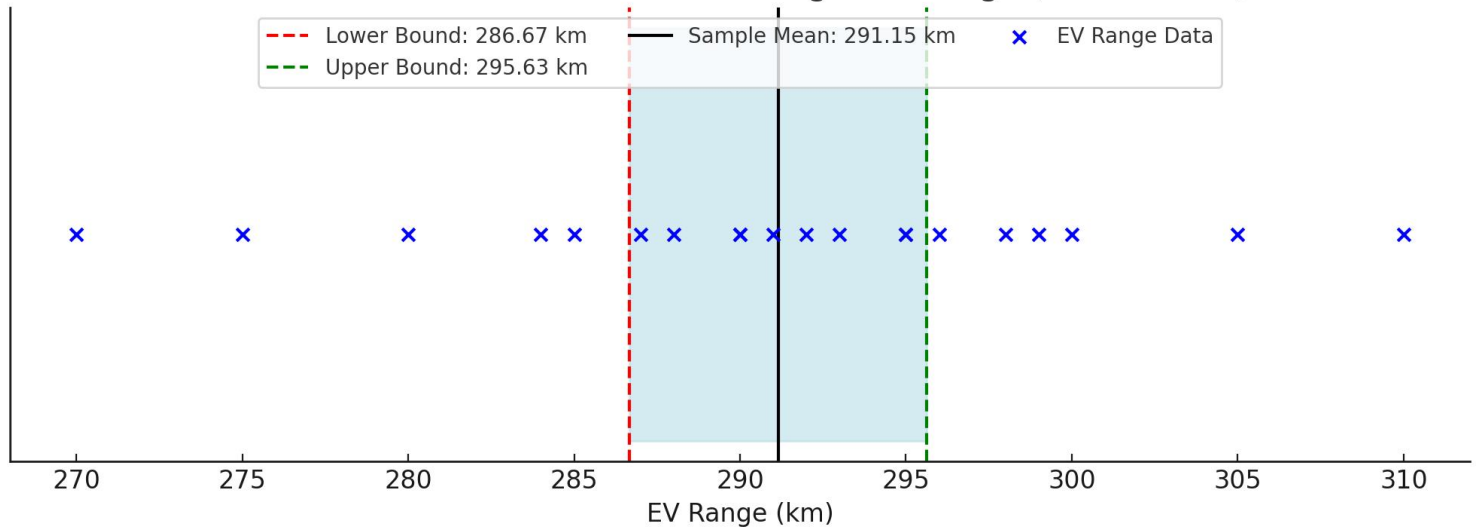
Black line: Sample mean (≈ 288.8 km)

Blue dots: Actual EV test results

Shaded area: The confidence interval range

This gives a clear picture of where the true mean range likely falls based on your 10-sample test.

95% Confidence Interval for Average EV Range (20 Vehicles)



Here's the updated 95% confidence interval after testing 20 electric vehicles:

New Sample Mean: 291.15 km

Margin of Error: ± 4.48 km

Confidence Interval:
[286.67 km, 295.63 km]

What Changed?

The confidence interval is narrower compared to when only 10 EVs were tested.

This happens because a larger sample size reduces uncertainty, giving us a more precise estimate of the true average EV range.

Question 5 – Confidence Interval for EV Failure Rate

Problem

Out of 150 EVs tested, 12 experienced charging station issues.

$$\hat{p} = \frac{12}{150} = 0.08$$

Step 1: Compute Margin of Error (Z-distribution)

$$z^* = 1.96 \text{ (for 95\% confidence)}$$

$$ME = z^* \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = 1.96 \cdot \sqrt{\frac{0.08 \cdot 0.92}{150}} \approx 0.0434$$

Step 2: Construct the Confidence Interval

$$CI = \hat{p} \pm ME = 0.08 \pm 0.0434 \Rightarrow [0.0366, 0.1234]$$

Interpretation

We are 95% confident that the true proportion of EVs likely to face charging station issues is between **3.66% and 12.34%**.

Impact of Sample Size

Sample Size	95% CI Range	Margin of Error
150 EVs	[3.66%, 12.34%]	$\pm 4.34\%$
300 EVs	[4.93%, 11.07%]	$\pm 3.07\%$
500 EVs	[5.62%, 10.38%]	$\pm 2.38\%$

Observation

Larger sample sizes lead to smaller margins of error and tighter confidence intervals. This increases the precision of the estimate.

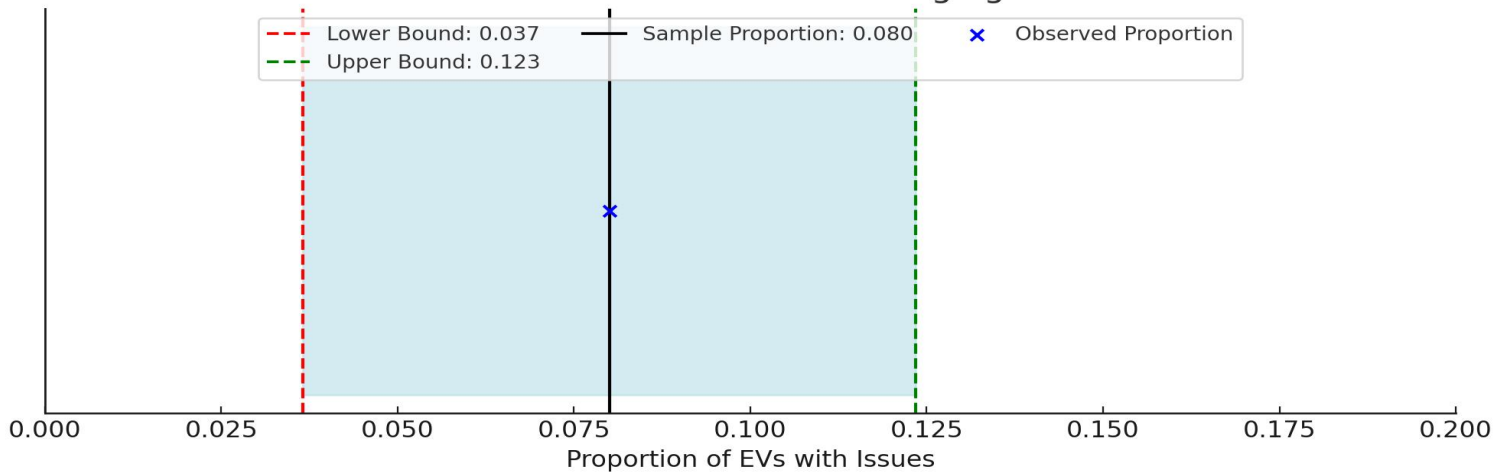
Impact of Confidence Level and Sample Size Together

Sample Size	Confidence Level	CI Range	Margin of Error
1000	95%	[6.32%, 9.68%]	$\pm 1.68\%$
1000	99%	[5.79%, 10.21%]	$\pm 2.21\%$
2000	95%	[6.81%, 9.19%]	$\pm 1.19\%$
2000	99%	[6.44%, 9.56%]	$\pm 1.56\%$
5000	95%	[7.25%, 8.75%]	$\pm 0.75\%$
5000	99%	[7.01%, 8.99%]	$\pm 0.99\%$

Observation

Higher confidence levels yield wider intervals to increase certainty. Larger sample sizes dramatically tighten the interval, giving more reliable estimates.

95% Confidence Interval for EV Charging Issue Rate



Here's the visual breakdown of the 95% confidence interval for EV charging issues:

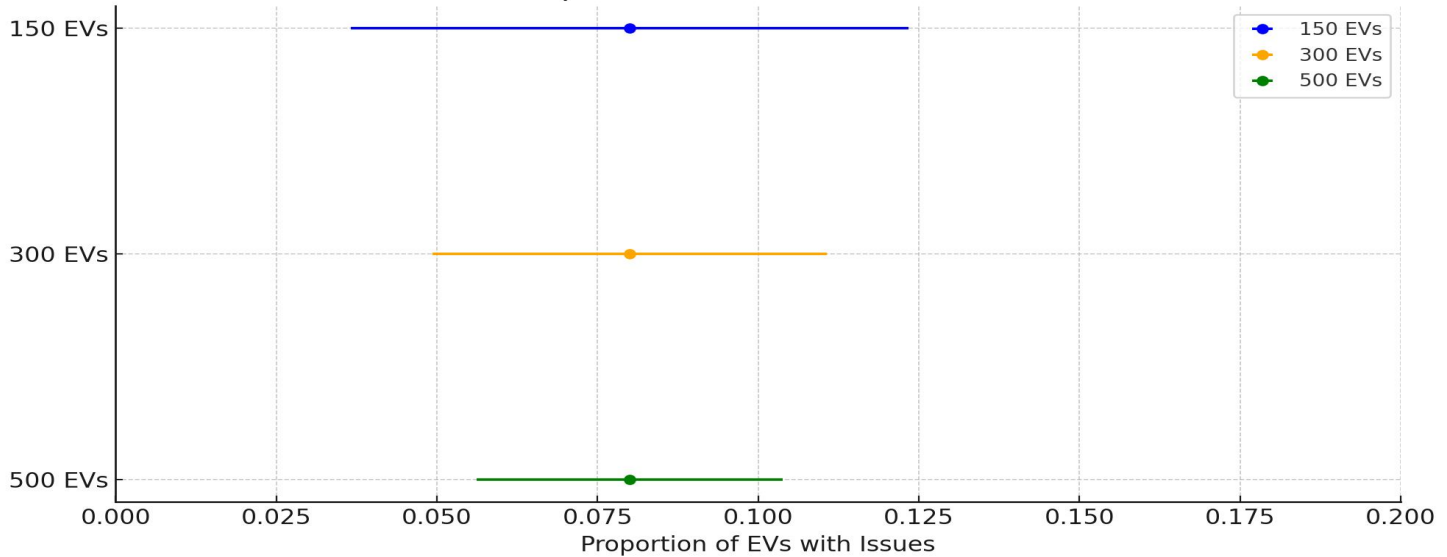
Black line: Observed issue rate (8%)

Red & Green dashed lines: Lower and upper confidence bounds

Shaded area: Likely range where the true issue rate falls (between ~3.66% and ~12.34%)

This gives a clear, visual sense of the uncertainty and reliability in estimating how often EVs might face charging station issues based on your sample.

Effect of Sample Size on Confidence Interval Width



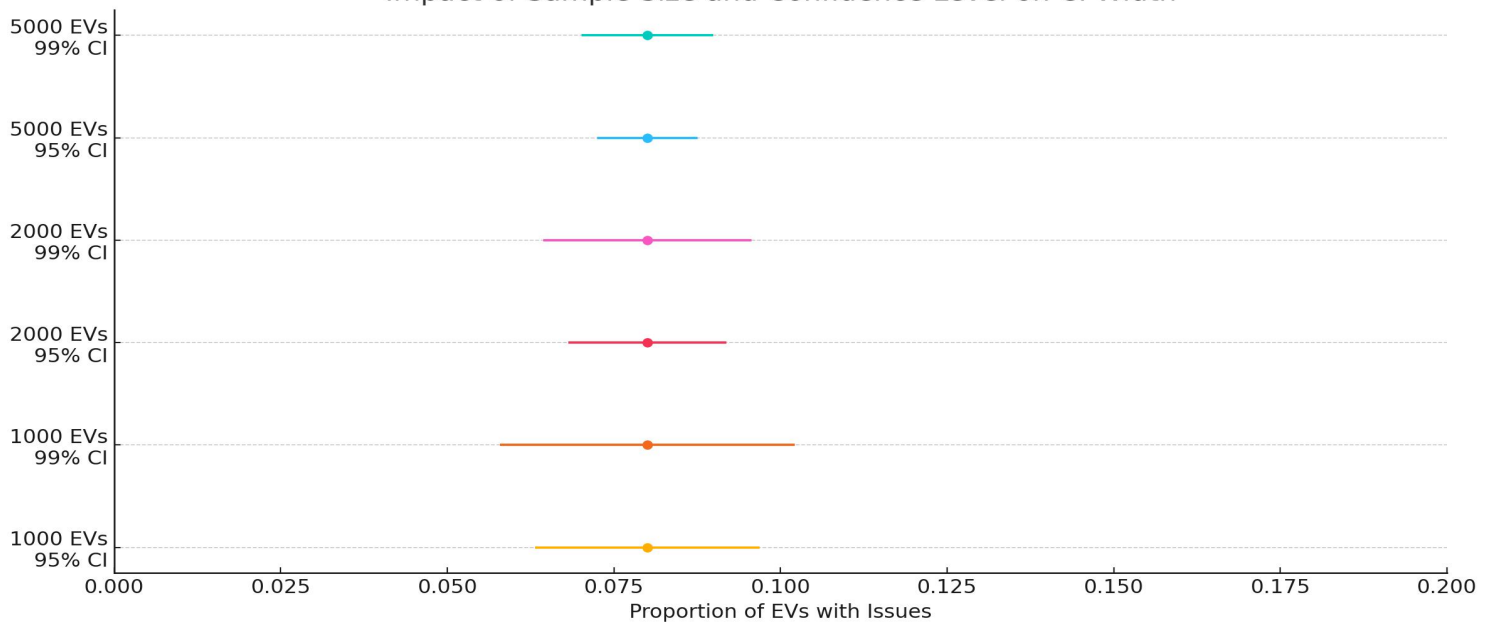
As the sample size grows:

The confidence interval becomes narrower.

Our estimate of the true issue rate becomes more precise.

We're better able to pinpoint the actual proportion of EVs likely to face issue

Impact of Sample Size and Confidence Level on CI Width



Takeaways:

Increasing the sample size greatly reduces the margin of error.

Raising the confidence level (e.g., from 95% to 99%) slightly widens the interval to increase certainty.

With 5,000 EVs, the estimate becomes very precise – great for national-level studies!

Part C – Interpretation & Decision Making

Q6. Interpretation Challenge

Incorrect Interpretation

“The 95% confidence interval for battery range is (310 km, 325 km), so there’s a 95% chance the true range is in this interval.”

Correct Interpretation

A 95% confidence interval means that if we were to repeat the sampling process many times, about 95% of the constructed intervals would contain the true mean battery range.

It is not about probability for one interval, but about the reliability of the method.

Key Insight

In a repeated sampling context, confidence intervals are designed to capture the population mean 95% of the time. For a given computed interval, the true mean is either in it or not—we don’t assign probabilities to fixed intervals.

Here's the visual to help explain confidence intervals intuitively:

Each horizontal line is a 95% confidence interval from a different sample.

The blue dashed line is the true (but unknown) mean battery range: 317 km.

Green lines are intervals that captured the true mean.

Red lines are intervals that missed it.

Key Insight:

About 95% of the intervals (roughly 24 out of 25) contain the true mean – that's what "95% confidence" really means!

You're not saying "this one interval has a 95% chance", but rather, "the process gives correct intervals 95% of the time."

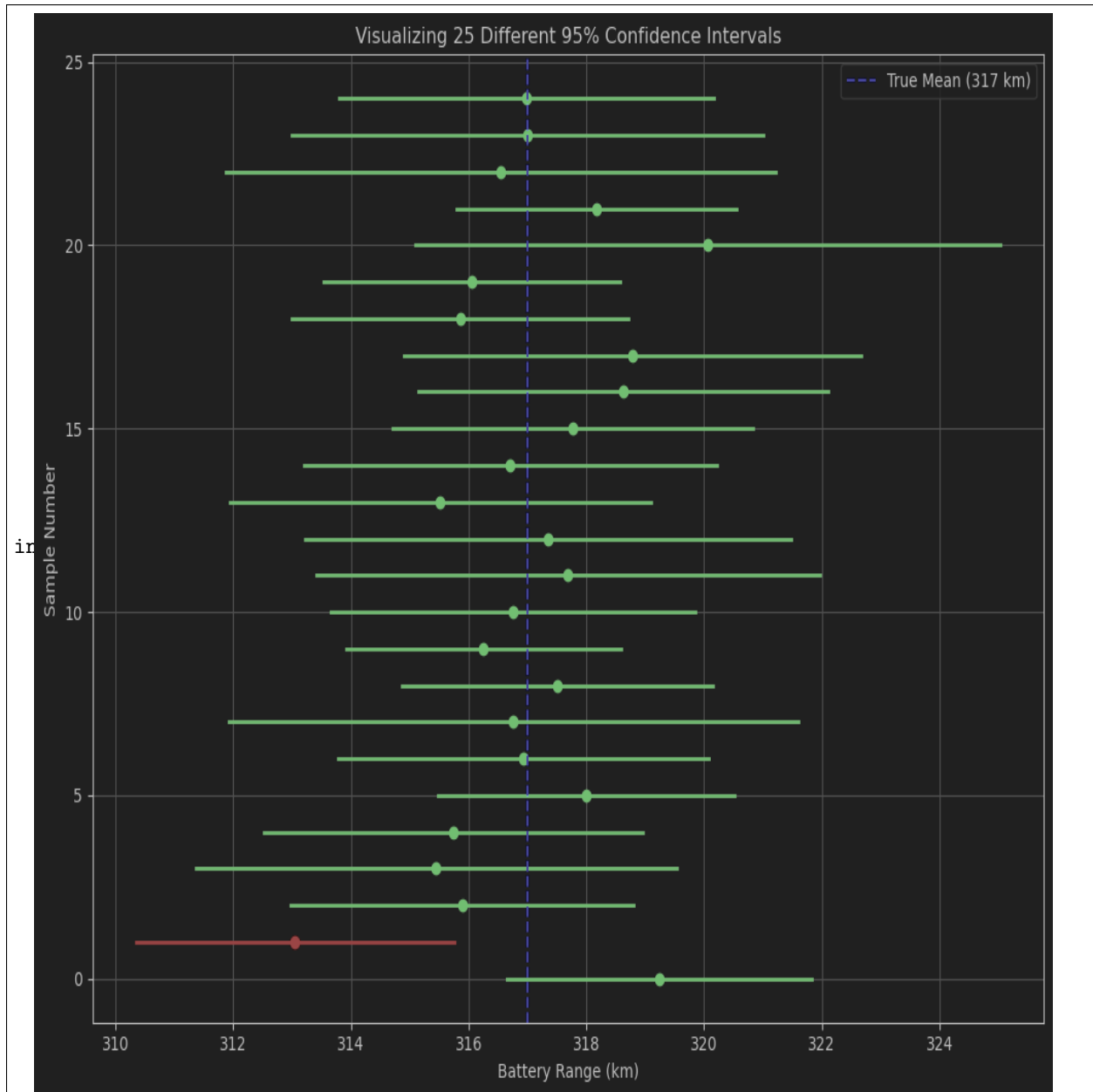


Figure 1: Visualizing 25 Different 95% Confidence Intervals

Q7. One-Sided vs Two-Sided Decision

Scenario

You are evaluating a new EV model that claims a minimum range of 320 km.

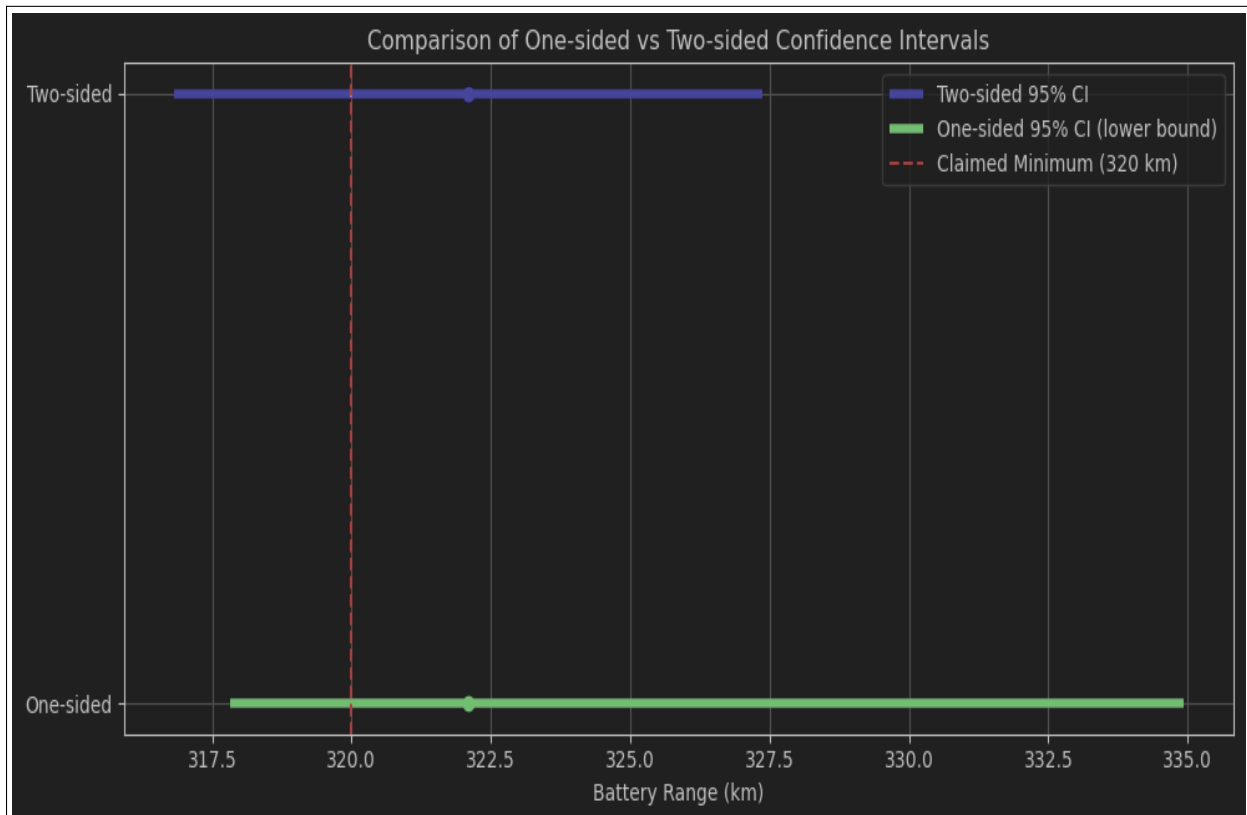
Use a One-Sided Confidence Interval

You want to know if the EV's range is **at least 320 km**. So, the focus is on testing whether the mean is not less than a threshold.

A **one-sided interval** is more appropriate here.

Two-Sided CI – When to Use Instead

Use a **two-sided CI** when you're trying to estimate or test for deviation in both directions — for instance, if you want to see if the mean is **different** from a standard, regardless of direction.



Here's the visual comparison:

Blue Line: A two-sided 95% CI, checking how far the mean might be on both sides.

Green Line: A one-sided 95% CI, focused only on ensuring the range doesn't fall below 320 km.

Red Dashed Line: The manufacturer's claimed minimum of 320 km.

Interpretation:

The two-sided CI is wider, but checks for any deviation.

The one-sided CI is tighter and directly answers your real question:

Figure 2: One-Sided vs Two-Sided Confidence Interval

Q8. Prediction Interval for Individual EV Range

Given:

- Sample mean range: $\bar{x} = 310$ km
- Standard deviation: $s = 12$ km
- Sample size: $n = 25$
- Confidence level: 95%

Prediction Interval Formula:

$$PI = \bar{x} \pm t^* \cdot s \cdot \sqrt{1 + \frac{1}{n}}$$

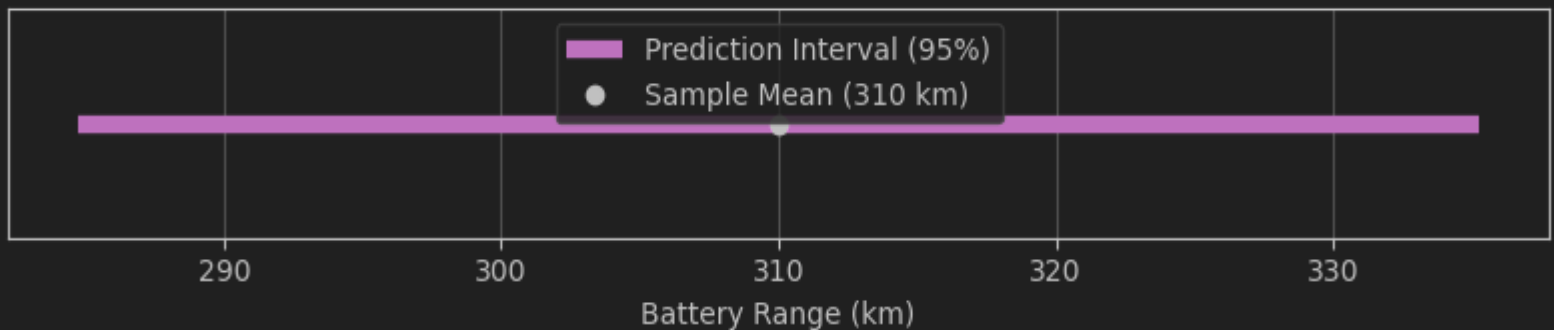
$$PI = 310 \pm t_{0.975,24} \cdot 12 \cdot \sqrt{1 + \frac{1}{25}} \approx 310 \pm 25.26$$

$$PI \approx (284.74 \text{ km}, 335.26 \text{ km})$$

Interpretation

This interval predicts where the range of a **single new EV** is likely to fall — not the average of many.

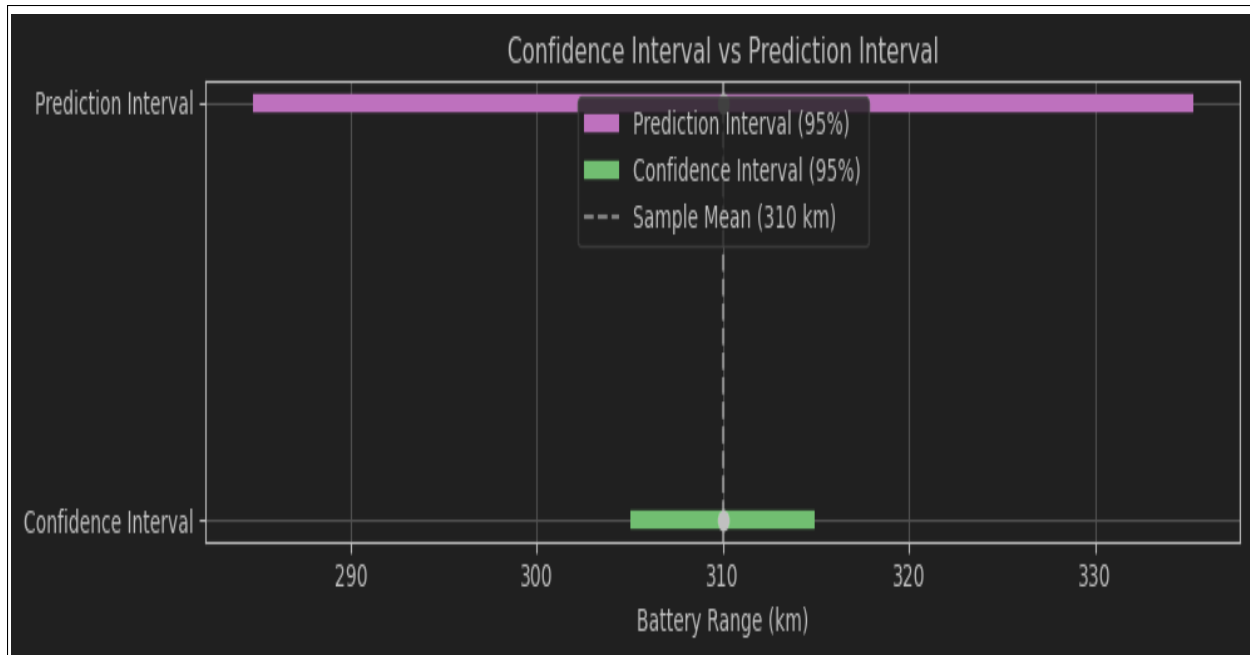
95% Prediction Interval for a New Individual EV



Interpretation:

This means a single new EV is likely to have a range between ~285 km and ~335 km.

It's wider than a confidence interval because it accounts for individual variation, not just uncertainty in the mean.



Here's the visual comparison:

Green Line: Confidence Interval (CI) for the mean – tighter, because it's about estimating the average range of EVs.

Purple Line: Prediction Interval (PI) for a new individual EV – wider, because it includes individual variability.

Black dashed line: The sample mean (310 km).

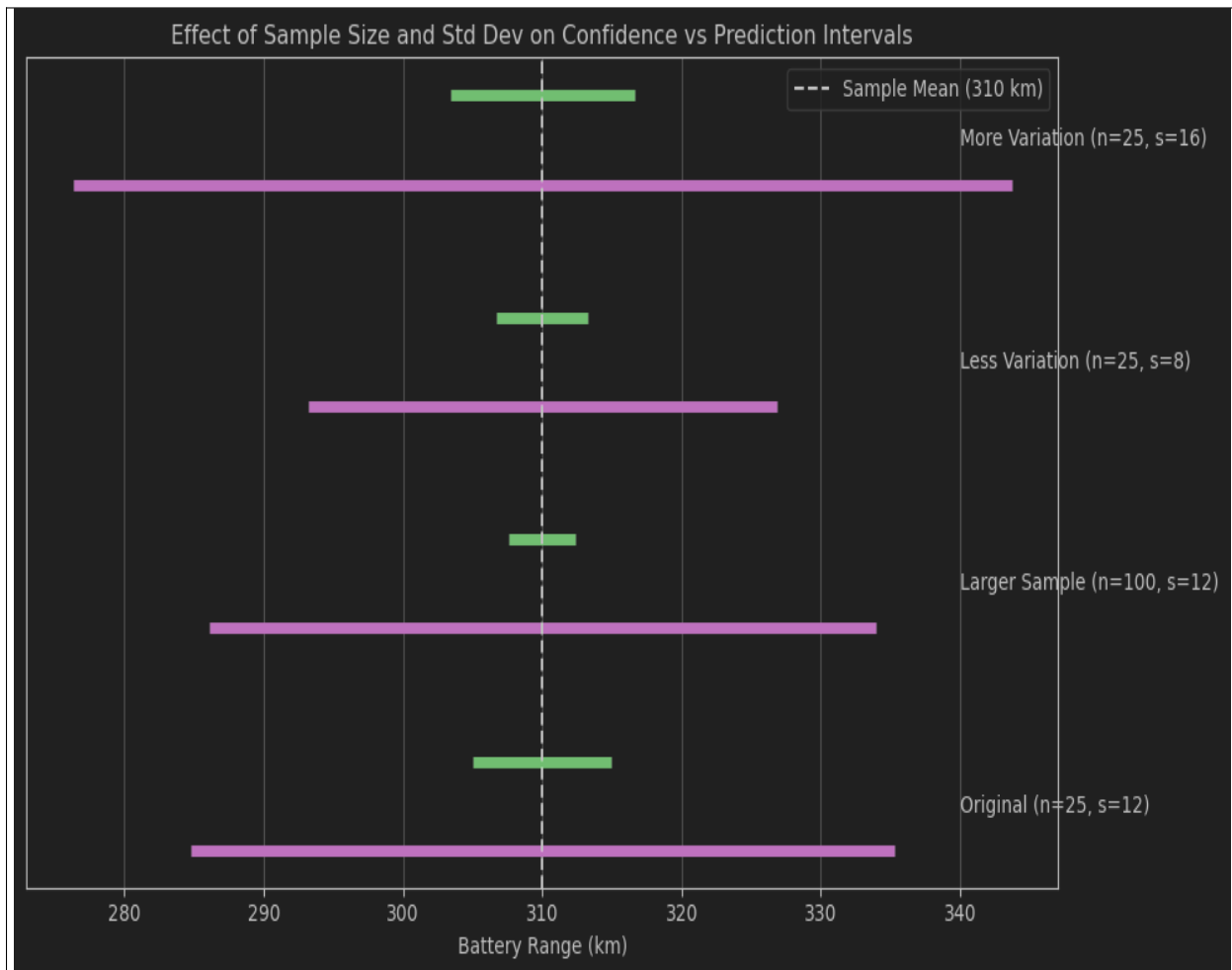
Key Takeaway:

Use the confidence interval to estimate how precise your average is.

Use the prediction interval when you care about what to expect from one future EV.

Figure 3: Confidence Interval vs Prediction Interval

Additional Comparisons:



Insight:

CIs shrink more rapidly with increasing sample size.

PIs are always wider, and less sensitive to sample size because they include individual-level variability.

Figure 4: Effect of Sample Size and Standard Deviation on Intervals

Summary Table

Scenario	CI Effect	PI Effect
Original (n=25, s=12)	Moderate width	Moderate width
Larger Sample (n=100)	CI narrows a lot	PI narrows slightly
Less Variation (s=8)	CI + PI both shrink	More precise
More Variation (s=16)	CI + PI both widen	Greater uncertainty

Part D – Engineering Application Challenge

Q9. Real-World Analysis – Comparing EV Models

Scenario

A manufacturer wants to compare two EV models (Model A and Model B). Data is collected from 20 vehicles of each model:

Metric	Model A	Model B	Unit
Sample Mean	295	310	km
Standard Deviation	10	8	km
Sample Size	20	20	–

Construct 95% confidence intervals and assess if one model significantly outperforms the other.

Step 1: CI Formula

$$CI = \bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

Step 2: Confidence Interval Calculations

Model A – 95% CI

$$\bar{x}_A = 295, \quad s_A = 10, \quad n_A = 20, \quad df = 19$$

$$t_{0.975, 19}^* \approx 2.093$$

$$CI_A = 295 \pm 2.093 \cdot \frac{10}{\sqrt{20}} \approx (290.32, 299.68)$$

Model B – 95% CI

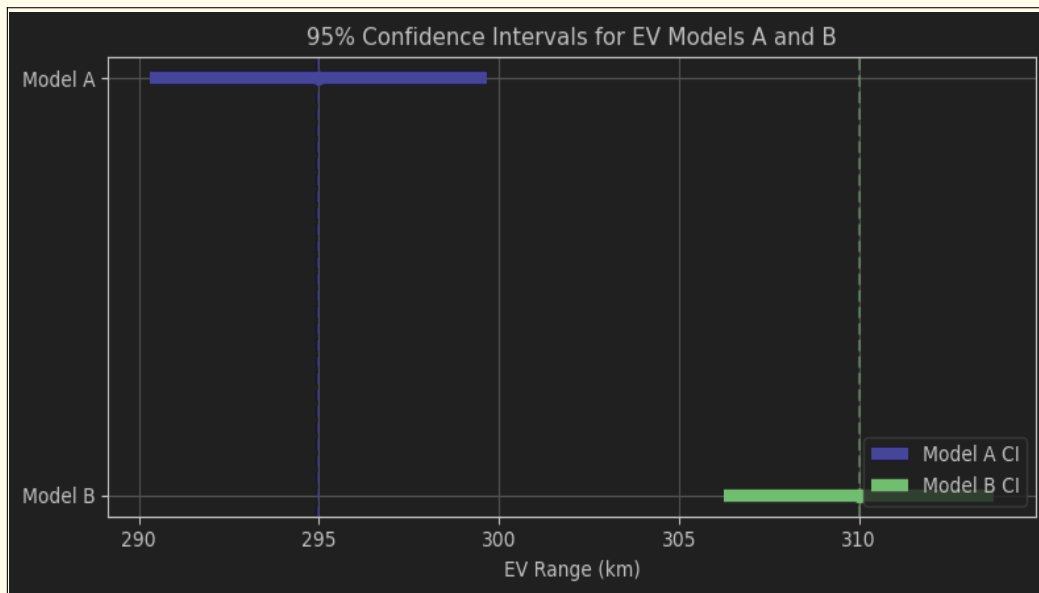
$$\bar{x}_B = 310, \quad s_B = 8, \quad n_B = 20, \quad df = 19$$

$$t_{0.975, 19}^* \approx 2.093$$

$$CI_B = 310 \pm 2.093 \cdot \frac{8}{\sqrt{20}} \approx (306.26, 313.74)$$

Visual Comparison

graph showing confidence intervals for Model A and Model B. Use horizontal lines with midpoint dots.



Model A (Blue): 95% CI from ~290.3 km to 299.7 km

Model B (Green): 95% CI from ~306.3 km to 313.7 km

☒ Key Observation:

The intervals do not overlap, giving strong evidence that Model B has a statistically higher average range than Model A.

Step 3: Interpretation

Overlap Check

$CI_A = (290.32, 299.68)$

$CI_B = (306.26, 313.74)$

Conclusion: There is **no overlap** between the two intervals.

Statistical Insight

When confidence intervals do not overlap, it strongly suggests that the difference in means is statistically significant. Thus, Model B **likely has a higher average range** than Model A.

Extension: Adding More Models

Additional Models Data

Metric	Model C	Model D	Unit
Sample Mean	305	300	km
Standard Deviation	12	11	km
Sample Size	20	20	–

CI Summary Table

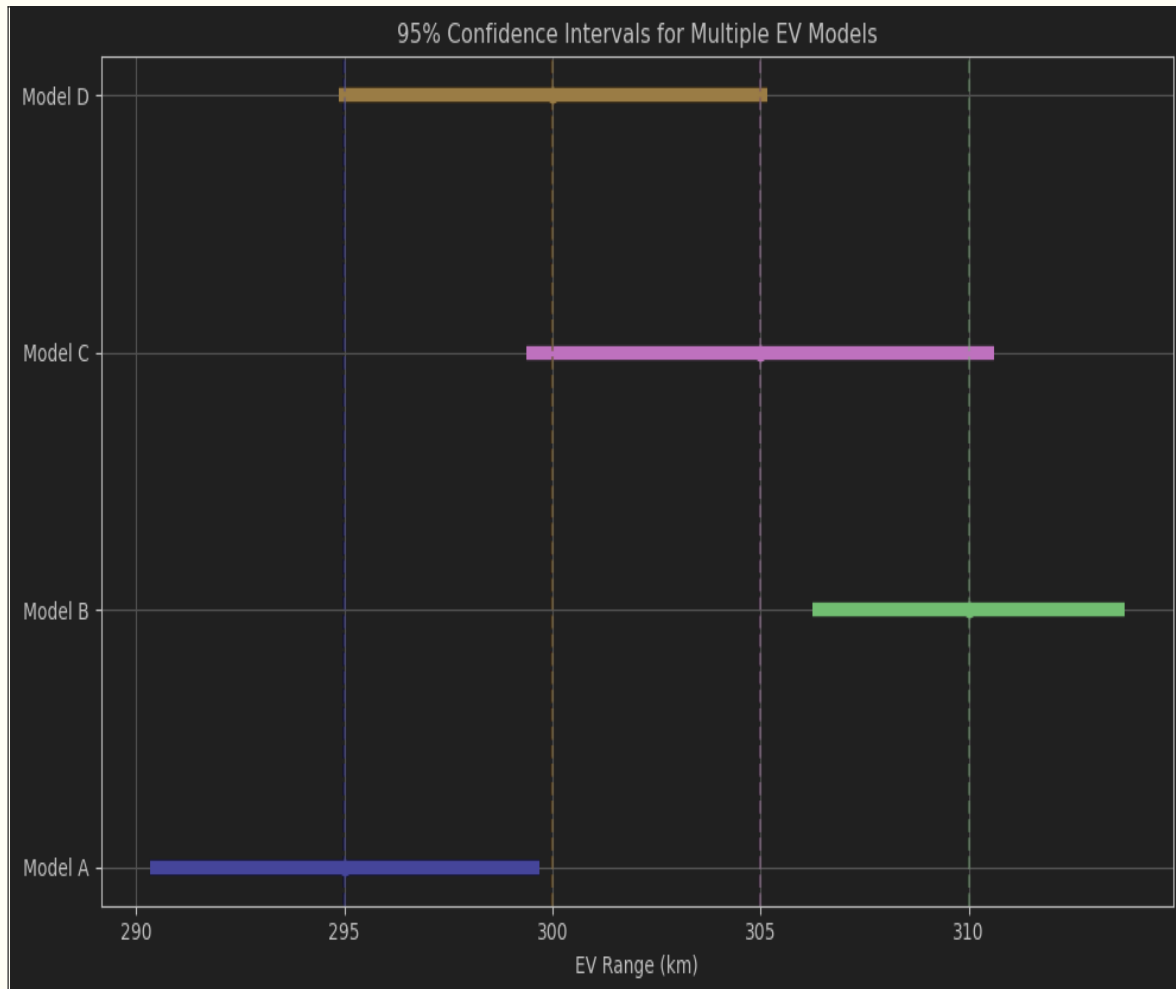
Model	95% CI
Model A	(290.32, 299.68)
Model B	(306.26, 313.74)
Model C	(298.38, 311.62)
Model D	(294.86, 305.14)

Insight

- Model B clearly outperforms all others in range. - Model C overlaps with both B and D, suggesting it's in-between. - Model A has the shortest average range, confirmed by non-overlapping CI with others.

Multi-Model CI Plot

horizontal CI plot for Models A, B, C, and D.



Model A: Shortest average range (~295 km)

Model B: Highest average range (~310 km)

Model C: Close to B, but more variation

Model D: Between A and C

Insights:

Model A vs B: Clear separation – B is better.

Model B vs C: Overlapping intervals → not a clear winner.

Model D overlaps with A and C → needs more data to confirm.

Part E – Reflection

Q10. How can Confidence Intervals Help Improve EV Reliability and Performance Benchmarks?

Concept Overview

Confidence intervals (CIs) help engineers and decision-makers understand not just the average performance of electric vehicles (EVs)—like range, charging time, or failure rate—but also how much **uncertainty or variability** is present in the data.

Key Benefits of Using Confidence Intervals in EV Engineering

Better Estimation of True Performance

- Confidence intervals give a **range** in which the true average performance is likely to lie.
- For example: Instead of saying “the average EV range is 310 km,” engineers can say:

We are 95% confident the true average range is between 305 km and 315 km.

Detecting Inconsistencies or Quality Issues

- **Wide intervals** may indicate inconsistency in production or reliability.
- **Narrow intervals** suggest a well-controlled and consistent manufacturing process.

Setting More Realistic Benchmarks

- CIs help set **tolerance limits** in design and deployment stages.
- Engineers can define **acceptable thresholds** under real-world variability.

Reducing Risk of Overpromising

- CIs avoid misleading “best-case” performance claims.
- This builds **consumer trust** and improves transparency.

Comparing Models with Statistical Rigor

- CIs enable **data-driven model comparisons**.
- If intervals don't overlap, it is strong evidence that one model performs significantly better.

Final Insight

Confidence intervals are powerful tools that promote precision, reduce risk, and support more reliable performance benchmarking in EV production and deployment. They bridge the gap between raw data and smart engineering decisions.

Visualization

concept map or infographic here showing how confidence intervals impact: production, testing, quality control, decision-making, and reliability.

